REVERSE ENGINEERING

AI Navigation System

Architecture, algorithms, data flow and redesign

REAL-WORLD CASE

Google Maps navigation — a conceptual reconstruction based on publicly documented system behavior.

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SIMATS ENGINEERING

CO5- ASSESSMENT TOOL 3 – Technical Presentation

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Code of Conduct:

I **B Santhosh Kumar/192511053** *(Name / Reg No) certify that this submission is my original work and that I have adhered*
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

B Santhosh Kumar
Signature of the student:

1. System Analysis Presentation

Present a reverse engineering analysis of an AI-based system, explaining its architecture, components, and working process using diagrams.

2. Architecture Reconstruction Demo

Prepare a technical presentation to reconstruct the architecture of a given software/AI system, highlighting module interactions and data flow.

3. Algorithm & Logic Identification

Present how you identified algorithms and decision-making logic in a reverse-engineered system, including methodology and tools used.

4. Data Flow & Processing Pipeline

Explain through a presentation the data flow, processing pipeline, and communication mechanisms in a reverse-engineered distributed system.

5. Enhancement & Redesign Proposal

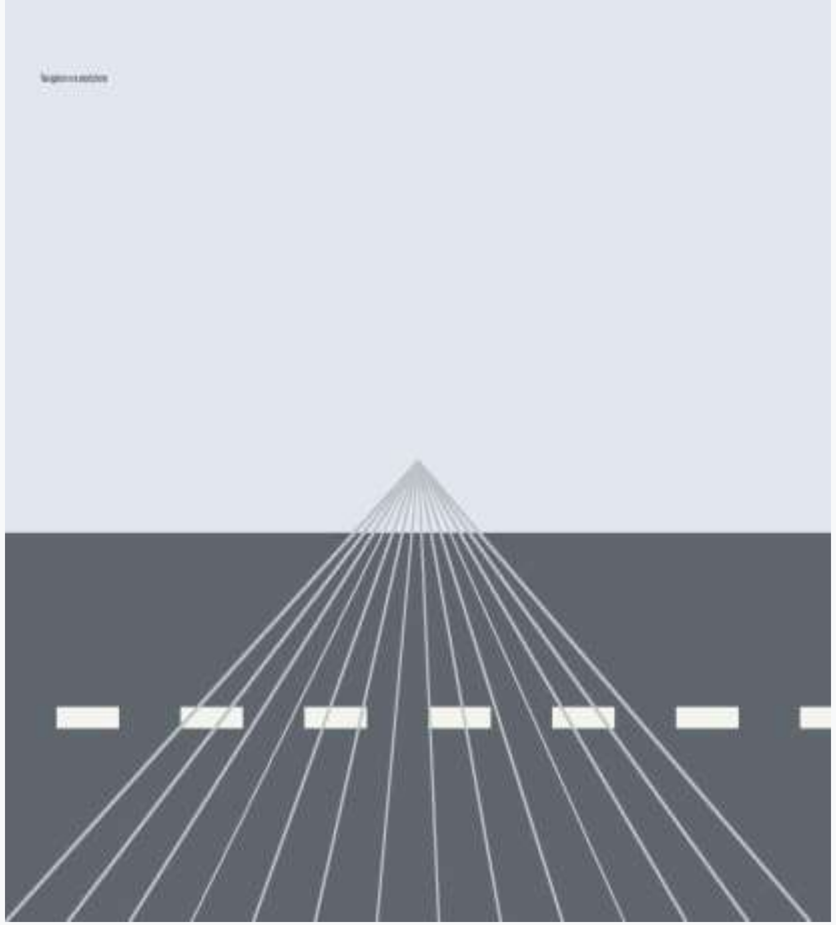
Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. System Analysis

What we can observe before looking inside the system



Inputs

GPS location
Destination
Road network
Traffic observations

Outputs

Route
ETA
Traffic condition
Rerouting / alerts

Main observation

The system combines live road conditions with historical patterns, predicts near-future traffic, then uses that information to select or change a route.

Objective

Infer the smallest architecture that explains these visible behaviors.

2. Reconstructed Architecture

A practical engineering view of the major modules

Mobile / Navigation App

GPS, destination, route display, voice guidance

Data Layer

Location signals, road graph, incidents, historical traffic

AI Prediction

Traffic forecasting and ETA estimation

Routing Engine

Candidate routes, constraints and route ranking

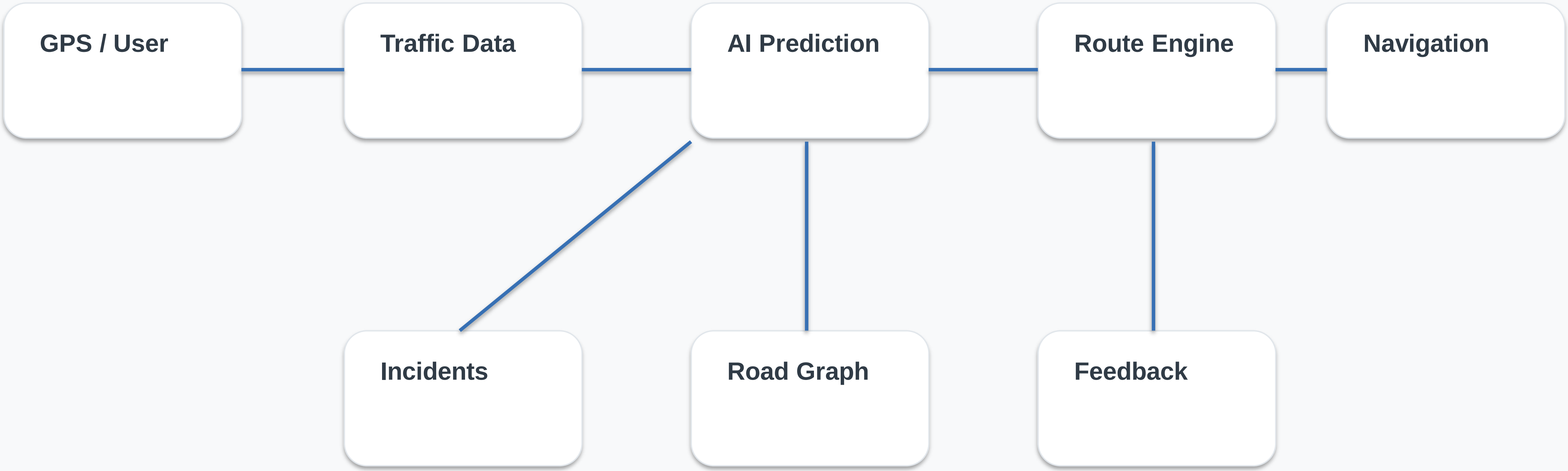
Response Layer

Best route, ETA, alerts and rerouting

Conceptual reconstruction — not Google's private source-code architecture.

3. Data Flow & Module Interaction

How information moves through the system



1. Collect
2. Predict
3. Generate routes
4. Rank
5. Deliver

The feedback loop matters: new road conditions can trigger another prediction and route decision.

4. Algorithm & Logic Identification

Algorithms inferred from the behavior and public technical evidence

Traffic Prediction

- Historical traffic patterns
- Live traffic observations
- Machine learning
- Graph Neural Network (publicly documented by Google DeepMind)
- Travel-time prediction

Route Selection

- Candidate route generation
- ETA comparison
- Road restrictions
- Road quality
- Directness / road size
- Alternative route when predicted congestion rises

Decision Logic

- IF route is restricted → reject
- IF predicted delay rises → compare alternatives
- IF alternative has lower cost → reroute
- ELSE → continue current route

Public technical evidence identifies a Graph Neural Network traffic-prediction component and a route-ranking stage.

5. Route Scoring Logic

A simple formula that explains the engineering trade-off

Illustrative comparison

Lower total cost → preferred feasible route

Route	ETA	Traffic	Cost
A	28 min	High	41
B	31 min	Medium	36
C	25 min	High risk	43

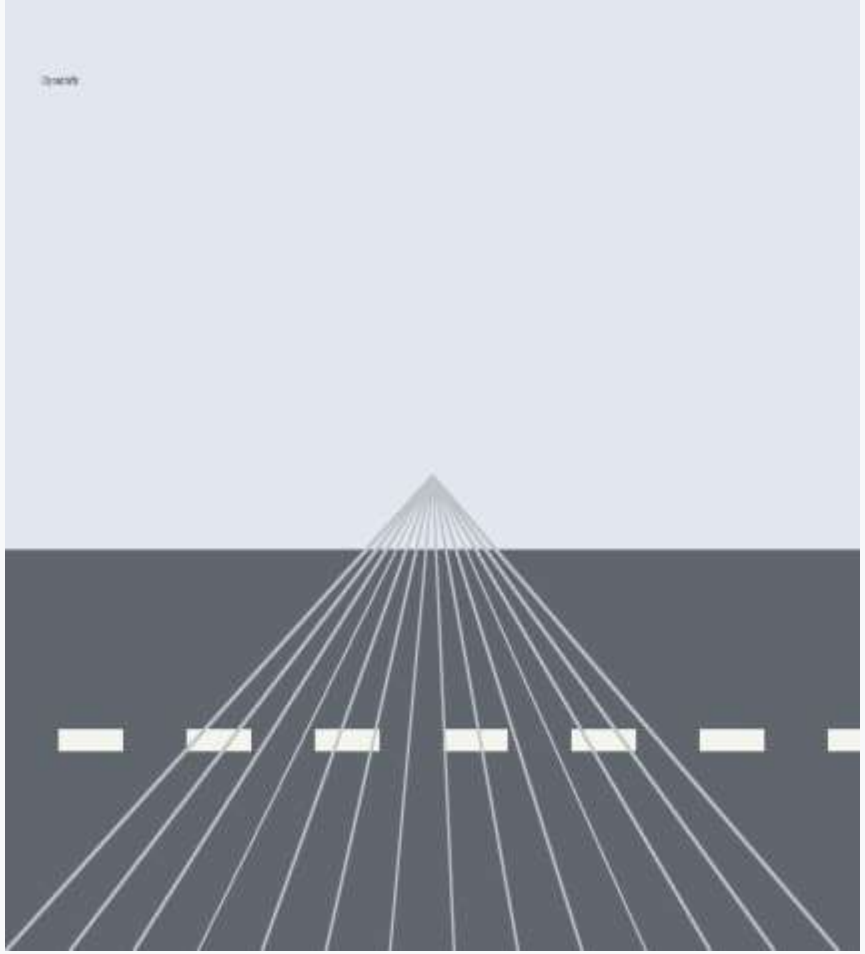
Values are illustrative, not Google Maps measurements.

Decision

Choose the lowest-cost route that satisfies road, safety and user constraints.

6. Real-World Example: Chennai Commute

A normal navigation decision, shown step by step



08:20 — **Start**
Driver enters destination.

08:24 — **Observe**
Traffic speed drops on current route.

08:25 — **Predict**
Model estimates additional delay.

08:26 — **Compare**
Routing engine checks alternatives.

08:26 — **Reroute**
Lower-cost route is recommended.

08:31 — **Update**
Fresh observations trigger another check.

This mirrors Google's publicly described example of predicting future congestion and rerouting a driver before the jam is reached.

7. Enhancement & Redesign Proposal

Improvements derived directly from the reverse-engineering findings

Problem Found

- Black-box decisions can be hard to explain.
- AI predictions may become less reliable during unusual traffic conditions.
- A delayed service can make a route decision stale.

Proposed Change

- Add prediction confidence.
-
- Separate safety / road constraints from ML.
-
- Add deterministic fallback routing.
-
- Store route-decision reason codes.

Expected Benefit

- More explainable routes
- Safer failure handling
- Easier testing
- Better monitoring
- Clearer debugging and governance

Design principle: use AI for prediction, but keep critical constraints deterministic and auditable.

8. Reverse-Engineering Method

How the architecture and logic were identified

Observe

Record inputs, outputs and edge cases.

Decompose

Separate UI, data, AI and routing responsibilities.

Hypothesize

Map behavior to likely algorithms and services.

Validate

Check whether the model explains multiple scenarios.

Document

Produce architecture, data-flow and decision diagrams.

Tools

Browser/app observation • API traces • Python • diagrams • logs

Validation

Scenario testing • boundary cases • timing checks • comparison of outputs

Ethical boundary: inspect only public behavior or systems you are authorized to test.

9. Conclusion

One reverse-engineering process connects all five assessment areas

Analyze

Understand visible inputs, outputs and responsibilities.

Reconstruct

Build the layered architecture and module interactions.

Identify

Infer algorithms and decision logic from evidence.

Trace

Follow data through prediction, routing and delivery.

Improve

Redesign for explainability, resilience and governance.

Final takeaway

Reverse engineering is not guessing the source code; it is creating the smallest technical model that explains the system's observed behavior.

Key sources: Google Maps 101 (2020); Google DeepMind traffic prediction (2020); Google Maps India AI updates (2023–2025).